# What Does 44 % Mean? Constant Baselines for Tooth-Level Caries Evaluation on Panoramic Radiographs

Unknown
*Author affiliations withheld pending submission*
*Unknown*
Unknown
unknown@unknown

***Abstract*—Automated screening of panoramic radiographs is almost always reported with tooth-level metrics: which teeth are carious. Such numbers are easy to compute, easy to compare, and—we argue—remarkably hard to interpret, because the literature rarely establishes what an uninformed system would score on the same images. We built a six-stage panoramic pipeline, validated it on four external hospitals, calibrated its confidences and gave it a principled way to decline, and then set out to answer a narrower question: what does its caries number actually mean? To find out, we scored it against progressively less informed opponents under one identical scorer—a 33 B-parameter vision–language model, an 8 B one, and finally a baseline that is not shown the radiograph at all. The comparison did not resolve the way we expected, and it did not resolve the same way inside and outside our development data. We report what separates a diagnostic system from an anatomical prior on this task, what does not, and which of our own results survive the distinction.**

***Index Terms*—panoramic radiography, dental caries, evaluation methodology, baselines, external validation, model calibration, selective prediction**

## I. Introduction

Consider a screening report. Against a panoramic radiograph it lists four tooth codes—16, 26, 36, 46—and marks them carious. A clinician reading that report has a reasonable first impression: the system has looked at the image and found something. The codes are specific. They are plausible. They are, often enough, correct.

Now consider a second program. It emits the same four codes for every radiograph it is ever shown. It contains no network, no features, no pixels— only a list. On a large enough set of panoramic images, it too will be right often enough to produce a respectable-looking number.

The distance between those two programs is the subject of this paper.

Panoramic radiographs (OPGs) are the most common extraoral radiograph in dentistry, and automated caries and periodontal screening on them is an active field [1]–[3]. Reported tooth-level accuracies cluster in a range that reads as clinically useful. What is much rarer is a statement of the floor: what the same metric would return for a system that does not perceive anything at all.

Two properties of this task make that floor unusually high, and both are anatomical rather than algorithmic. First, caries is not uniformly distributed across the dentition—it concentrates heavily in molars. Second, tooth-level metrics ask a *set-membership* question: which teeth, from a fixed vocabulary of 32 codes. A question of that shape can be answered from anatomy alone, without reference to any particular patient. Together these mean that a fixed answer inherits a substantial score by construction.

The natural response is to check. We surveyed 49 task-relevant papers from our own review of the area; **two** report validation on data from an institution other than the one that produced their training set, and **none** report a constant, image-blind, or otherwise uninformed baseline. The floor is, in this literature, essentially unmeasured.

This paper is our attempt to measure it on our own work rather than anyone else's. We first built a system worth testing—a six-stage pipeline performing tooth detection, FDI numbering, caries classification and periodontal bone-loss classification, evaluated with cluster-bootstrapped confidence intervals, calibrated probabilities, a pre-registered abstention threshold, and external validation across four hospitals it never trained on. We then constructed a ladder of opponents, each given strictly less information than the last, and scored every rung with one scorer on one set of images.

We report the outcome in full, including the parts that do not favour our system, and including a disagreement between our development data and our external data that we did not anticipate and cannot yet explain. Our contributions are:

1) A **constant-baseline protocol** for tooth-level OPG evaluation, together with what it reveals when applied across two unrelated metric families and five hospitals.
2) A **vision–language-model baseline with the controls that make it interpretable**—including what these models answer when shown an image containing no anatomy whatsoever.

3) **External validation on four hospitals**, reported per site, with the spread that pooling conceals.
4) **Calibration and selective prediction** results, each with a control isolating the mechanism responsible.
5) An honest characterisation of the pipeline itself, including one branch that does not survive domain shift and a headline number that is bounded by an earlier stage.

We are explicit about what we do not claim. We do not assert that published caries results are incorrect. We argue something narrower and, we think, more useful: that a tooth-level number reported without a floor cannot be distinguished from an anatomical prior—a statement about *evidence*, not about anyone's system.

## II. Related Work

### A. Detection and classification on panoramic radiographs

Deep learning on OPGs has converged on a familiar decomposition: detect teeth, then classify each one. Detector choice is well studied; a head-to-head evaluation on dental panoramics reports YOLOv11 outperforming its successor generation for multi-class dental detection, and the same study observes an asymmetry that matters here—performance is strong on well-defined objects such as crowns and implants and markedly weaker on subtle radiolucent pathology including caries and bone loss [4]. YOLO-family detectors are widely applied to this data more generally [5]–[7]. Caries and periodontal classification specifically have been approached with dual-network frameworks [1], dedicated convolutional classifiers [3], and contrastive feature-pyramid designs [2], alongside broader detection and segmentation work [8], [9].

### B. Tooth numbering

FDI numbering is typically reported at very high accuracy—one recent system reports 0.9988 for tooth numbering [10], and multi-stage segmentation-and-numbering pipelines report comparable figures [11]. Such numbers are frequently obtained on *pre-cropped* teeth, where the task is a 32-way classification of an image already known to contain exactly one tooth, correctly framed. That is a materially easier problem than numbering under detector output, where teeth may be missed, duplicated or clipped. We report both framings and find the gap between them large.

### C. Vision–language models in dentistry

Recent work applies general-purpose and domain-adapted VLMs to dental imaging [12]–[15]. These systems are attractive because they need no task-specific training, and are difficult to evaluate for the same reason: a fluent, confident, correctly formatted answer is not evidence of perception. We adopt a control from the broader shortcut-learning literature [16]: present the model with an input containing none of the signal it purports to read, and observe whether its behaviour changes.

### D. External validation and the missing floor

External validation remains uncommon. In our review of 49 task-relevant papers, two evaluate on an outside institution. Where it is done, degradation is often severe—one reported caries detector falls from 95.06 mAP on its own data to 61.83 on a public benchmark [17]. Independent work on periodontal bone loss, evaluated against five clinicians, reports weighted F1 of 0.29 with severe-class F1 of zero [18], a reminder that in-domain figures can be a poor guide to behaviour elsewhere. To our knowledge, no work in this area reports an image-blind baseline, which is the gap this paper addresses.

## III. System

The finding in this paper is only interesting if the system it is measured against is a serious one. We therefore describe the pipeline before the evaluation that tests it.

### A. Architecture

The pipeline maps an OPG to a structured per-tooth record in six stages. **Stage 1** detects teeth with a YOLOv11l segmentation model [19]. **Stage 2** assigns each detection a two-digit FDI code using a ResNet18 [20] head that consumes the tooth crop together with four positional features, resolved to a one-to-one assignment across the mouth by the Hungarian algorithm [21]. **Stages 3 and 4** classify caries and periodontal bone loss per tooth using independent ResNet18 branches, each arranged as a coarse-to-fine cascade. **Stages 5 and 6** fuse the per-stage outputs on the FDI key and emit the record.

The whole system totals approximately 25 M parameters and runs at 2.39 s per image on CPU with no GPU present. This figure becomes relevant in Section V.

### B. Data

Tooth detection and bone loss use the Tufts Dental Database [22] (1,000 OPGs) under a frozen 700/150/150 split. Caries uses DENTEX [23]. FDI numbering trains on both, with a third public corpus used for an ablation.

All splits are **image-level**. Two crops from one radiograph are not independent observations, and a tooth-level split would place the same mouth in both training and test data.

Two labelling conventions were established by measurement rather than assumed, because a silent error in either would have trained a plausible but wrong model. Tufts numbering is left–right **mirrored** relative to Universal notation (Tufts "1" corresponds to FDI 28, not 18), confirmed three independent ways. The third corpus was checked the same way: using the two-corpus numbering head as an independent arbiter over 915 teeth, agreement was **80.3 %** reading its labels as written against **1.7 %** reading them mirrored.

### C. External data

Four hospitals from an independent Chinese cohort (denoted test1–test4) serve as locked external test data: never trained on, never tuned against, and evaluated a bounded and recorded number of times.

### D. Crop geometry and pre-registration

Classifier inputs are expanded crops resized to $224 \times 224$. Expansion is fixed per branch and identical across corpora—symmetric $1.5\times$ for numbering, $1.5\times$ width by $2.0\times$ height for bone loss, since teeth are roughly three times taller than wide and horizontal over-expansion bleeds into neighbouring teeth.

Operating points are fixed *a priori* and reported where they fall rather than tuned against evaluation data. One consequence is reported here rather than omitted: the detector's mask-quality bar was pre-registered at 0.65 and **missed**, at 0.638. The bar was not moved after the fact; the model was adopted for its bounding boxes alone by an explicit, dated decision, and the mask-dependent component it was intended to enable remains unwired.

## IV. Evaluation Protocol

Having built the system, we wanted to know what its caries number meant. The protocol below is the paper's methodological core.

### A. The ladder

Four systems answer one question—*which teeth in this radiograph show caries?*—on the same 164 leakage-free DENTEX images, scored by one scorer per metric family. Each rung is given strictly less information than the one above.

1) **Pipeline**: the six stages of Section III.
2) **VLM (33 B)**: Qwen2.5-VL-33B-Instruct [24], zero-shot.
3) **VLM (8 B)**: the same family at 8 B.
4) **Constant**: a fixed set of FDI codes emitted for every image. **The radiograph is never read.**

### B. Fairness

A rigged baseline is worse than no baseline. The VLMs are given the FDI numbering scheme in the prompt, told explicitly that teeth may be absent, and parsed permissively, so that neither notation nor formatting is scored as clinical error. Our pipeline receives no corresponding assistance. Every affordance favours the opponent.

The constant is fitted greedily on 78 leakage-excluded images that are disjoint from the scored set—overlap asserted to be zero at runtime—and then frozen. We also report a variant fitted directly on the test set; it is labelled an oracle throughout and is not claimable.

### C. Metrics

We use two metric families, chosen to fail differently.

**FDI-set F1** treats each image as a set-prediction problem: micro-averaged precision, recall and F1 over predicted versus annotated FDI codes.

**Lesion-localisation agreement** asks, of every expert-annotated lesion, whether a tooth was flagged that spatially overlaps it. It requires spatial correspondence, which set membership does not. Because it has no negative class by construction—a deliberate property, since our external cohort lacks healthy controls—it is a recall-style measure, and flagging more teeth can only raise it. We therefore always report it alongside the predicted-positive match rate and the number of teeth flagged, and we include an upper-bound arm that flags *every* detected tooth.

TABLE I
TOOTH-LEVEL CARIES PREDICTION, 164 LEAKAGE-FREE DENTEX IMAGES, 477 AFFECTED TEETH. ONE SCORER FOR ALL ARMS.

| System | P | R | F1 |
|---|---|---|---|
| **Constant** (image never read) | 0.218 | 0.600 | **0.320** |
| Our pipeline | 0.235 | 0.390 | 0.294 |
| Qwen2.5-VL-33B, zero-shot | 0.250 | 0.243 | 0.246 |
| Qwen2.5-VL-8B, zero-shot | 0.248 | 0.107 | 0.149 |
| *Oracle constant (test-fitted)* | — | — | *0.330* |

TABLE II
LESION-LOCALISATION AGREEMENT, SAME IMAGES, 479 ANNOTATED LESIONS. TEETH FLAGGED IS REPORTED BECAUSE AGREEMENT ALONE REWARDS FLAGGING MORE.

| System | Agreement | Teeth flagged | Match rate |
|---|---|---|---|
| **Constant** | **0.488** | 859 | 0.272 |
| Our pipeline | 0.443 | 755 | 0.281 |
| *All teeth (ceiling)* | *0.722* | *3,344* | *0.103* |

### D. Controls

For the VLMs, we present a **blank grey image** and a **random-noise image** under the identical prompt. Neither contains a tooth. For selective prediction, we **shuffle the confidences**, breaking the link between confidence and correctness while preserving both marginals; a gain that survives shuffling is an artefact of discarding items rather than of ranking them.

## V. Results

### A. The ladder, on development data

Fig. 1 summarises the comparison; Table I gives the set-prediction result exactly.

The ordering is inverted with respect to the information each system receives. The best-performing entry is the one that never sees a radiograph.

A single metric behaving oddly is a metric artefact. We therefore repeated the comparison under lesion localisation, which demands spatial correspondence (Table II).

The constant leads again, by **4.59 percentage points**, 95 % CI $[+0.60, +8.89]$ under a paired bootstrap over images. It does not achieve this by indiscriminate flagging: it flags 14 % more teeth than the pipeline at an essentially identical match rate. Two metric families designed to measure different things therefore agree.

### B. The ladder, on external data

We then ran the same comparison on four hospitals the system had never seen (Table III).

**The development-set result does not replicate.** Pooled, the difference is $-0.89$ percentage points with 95 % CI

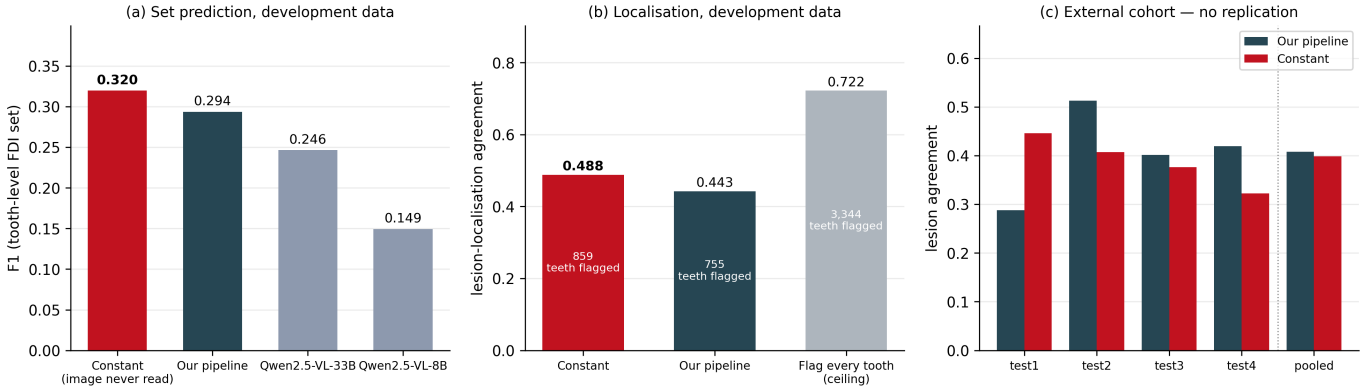



Fig. 1. The constant-baseline argument. **(a)** Set-prediction F1 on 164 leakage-free development images: the ordering is inverted with respect to the information each system receives, and the best-scoring entry never reads a radiograph. **(b)** A second, unrelated metric requiring spatial correspondence returns the same ordering; teeth flagged is annotated inside each bar because agreement is recall-shaped, and the flag-everything arm shows what saturating the metric costs. **(c)** The external cohort, where the result does *not* replicate: the difference swings 26 points between sites and vanishes when pooled. Every value is read from the saved evaluation artefacts at render time rather than transcribed.

TABLE III
EXTERNAL VALIDATION, FOUR HOSPITALS, 337 IMAGES, 1,013 LESIONS. CARIES LESION AGREEMENT BY CENTROID CONTAINMENT.

| Hospital | Lesions | Pipeline | Constant | Δ |
|---|---|---|---|---|
| test1 | 316 | 0.288 | **0.446** | +15.8 |
| test2 | 349 | **0.513** | 0.407 | −10.6 |
| test3 | 162 | 0.401 | 0.377 | −2.5 |
| test4 | 186 | 0.419 | 0.323 | −9.7 |
| **Pooled** | 1,013 | 0.408 | 0.399 | −0.89 |

$[-4.88, +3.01]$: the two systems are statistically indistinguishable, with the pipeline nominally ahead by an amount that is indistinguishable from noise.

The per-site spread is the more useful observation. The difference swings from $+15.8$ points at test1 to $-10.6$ at test2—a 26-point range across four hospitals of a single cohort—and the pooled figure conceals this entirely. Notably, test1 is simultaneously the pipeline's worst site and the constant's best. We do not have an explanation for that, and we report it unexplained rather than omit it.

Across all five hospitals, then, the defensible claim is *non-separability*: a purpose-built pipeline and an image-blind prior cannot be reliably told apart by these metrics. That is weaker than the development-set result alone would support, and it is the claim the full evidence sustains.

### C. What the constant cannot do

A constant answers a set-membership question. It cannot perform tasks whose output is not a fixed list, and the pipeline's results on those tasks are correspondingly meaningful.

**Tooth numbering.** End-to-end FDI-exact accuracy is 58.3 % (546/936 teeth), 95 % CI [53.9, 62.8]. This figure is **detection-bound** and we report it as two numbers: 29.4 % of ground-truth teeth are never detected, and of the 661 teeth Stage 1 does find, Stage 2 numbers **82.6 %** correctly. Residual errors are largely benign: of 115 incorrect codes, 89 (77.4 %) are off-by-one within the correct quadrant, and only 16 of 661 detected teeth (2.4 %) receive a quadrant-level error. A constant cannot attempt this task at all.

**Calibration.** The caries gate's confidences are not probabilities: expected calibration error is 0.136 and the model is systematically overconfident. Temperature scaling [25] made calibration *worse* (0.152), indicating the miscalibration is structural rather than a single scale factor; isotonic regression [26] reduced ECE to **0.043**, a 68 % improvement, and is applied to confidences only, leaving every verdict byte-identical.

**Selective prediction.** Declining the least-confident quarter of cases raises numbering accuracy from 0.826 to **0.885** ($+5.91$ pp, CI $[+3.88, +7.68]$) and caries-gate accuracy from 0.797 to **0.940** ($+14.25$ pp, CI $[+11.32, +16.63]$). Under the shuffle control both gains collapse to $+0.03$ and $+0.00$ percentage points respectively, establishing that the improvement comes from the confidence ranking and not from discarding items. The pre-registered threshold of 0.30 lands at 90.8 % coverage and 0.847 accuracy, within 0.2 points of the empirical 90 %-coverage operating point—a value chosen before these data were seen.

One caveat must accompany any coverage figure here: coverage is measured over *detected* teeth. Retaining 75 % of detections corresponds to 75 % of 70.6 %, or 53.0 % of the teeth actually present.

### D. Vision–language model behaviour

Few-shot prompting, with exemplars drawn strictly from leakage-excluded images, degrades the 8 B model monotonically: F1 falls from 0.149 zero-shot to 0.066 at two shots and 0.034 at four. The mechanism is abstention rather than error— empty answers rise from 56 of 164 images to 123 of 164—so exact-set match simultaneously *rises*, because 16 images contain no caries and an empty answer scores an exact

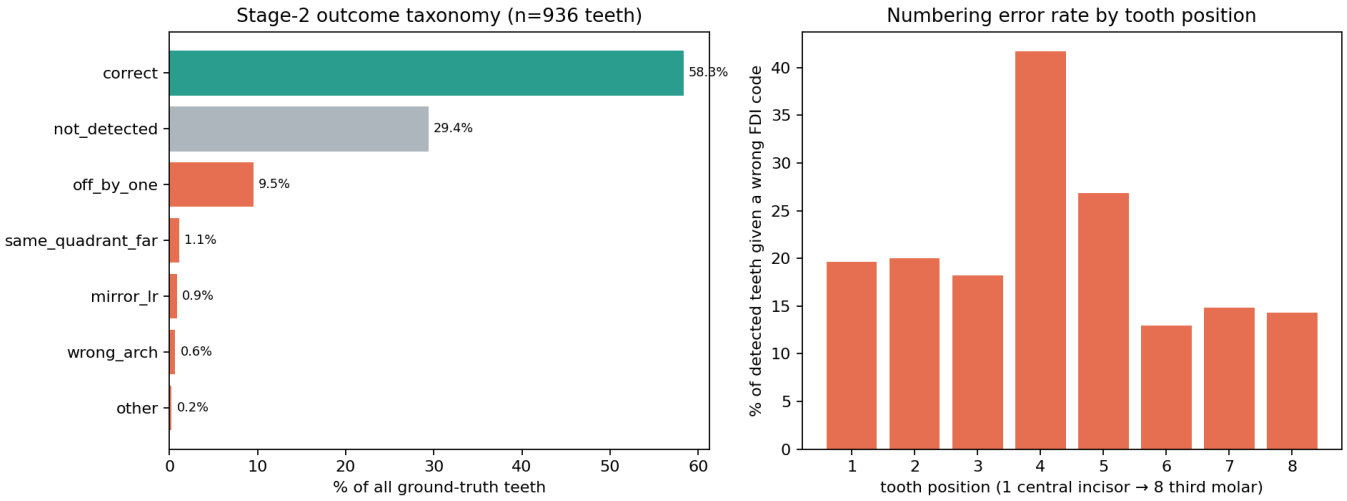



Fig. 2. Where the numbering stage loses teeth. The largest single category is teeth the detector never proposes, which no numbering model can recover; among the codes that are wrong, most are off-by-one within the correct quadrant.

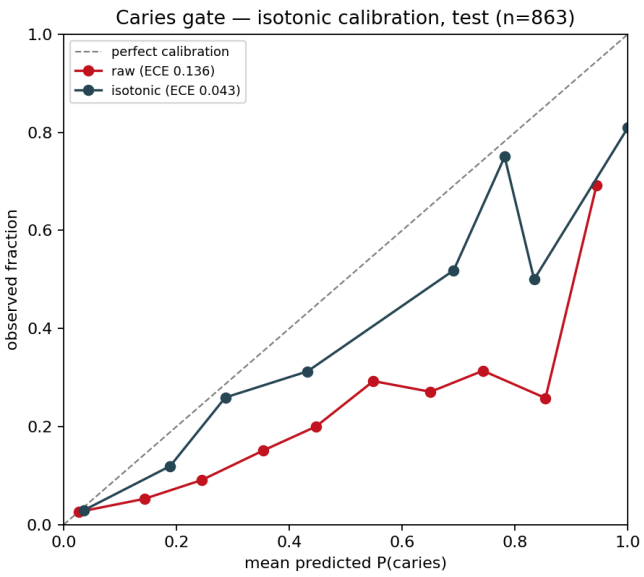



Fig. 3. Reliability of the caries gate before and after isotonic regression. The uncorrected model is systematically overconfident. The correction is applied to confidences only, leaving every verdict unchanged.

match on each. Reported alone, that column would be true and thoroughly misleading.

The control is more direct. Presented with a **blank grey image** and with **random noise**, both models return confident, correctly formatted, dental-specific answers; for one model the response to pure noise is identical to its response to a real radiograph. Whatever these outputs measure, it is not perception of the image in front of them.

The efficiency comparison survives all of the above and is worth stating plainly: 33 B parameters and 70 GB of accelerator memory are outperformed by approximately 25 M

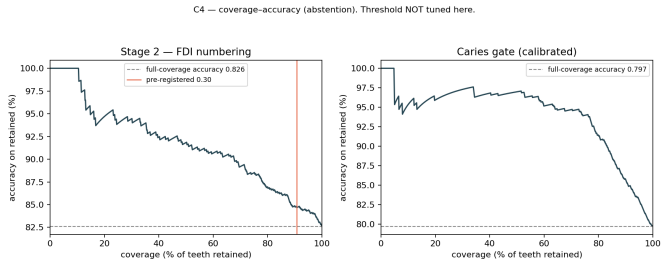



Fig. 4. Coverage–accuracy for tooth numbering (left) and the caries gate (right). Accuracy on retained predictions rises as the least-confident cases are declined. The vertical line marks the pre-registered threshold of 0.30, located on the curve rather than tuned against it.

parameters running on a CPU.

### *E. Where the pipeline fails*

The bone-loss branch reaches 0.923 macro-F1 in-domain on 1,558 validation crops at an untouched 4 % prevalence (sensitivity 83.05 %, specificity 99.53 %) and **does not survive domain shift**, firing on 0.7 % of DENTEX teeth and reaching only 1.4–5.6 % affected-tooth sensitivity across the four external hospitals. A detector change that moved numbering by +12.0 points and caries agreement by +7.7 moved bone loss by nothing, which falsifies the hypothesis that a shared upstream cause explains all three. We report this as a measured negative result rather than a component.

## VI. Discussion

### *A. Why the floor is high, and why that is not a criticism of any model*

Caries in these cohorts is anatomically concentrated: molars carry most of the disease. A tooth-level decision is a set-membership question over a fixed vocabulary of 32 codes, and set membership inherits that concentration directly. A fixed

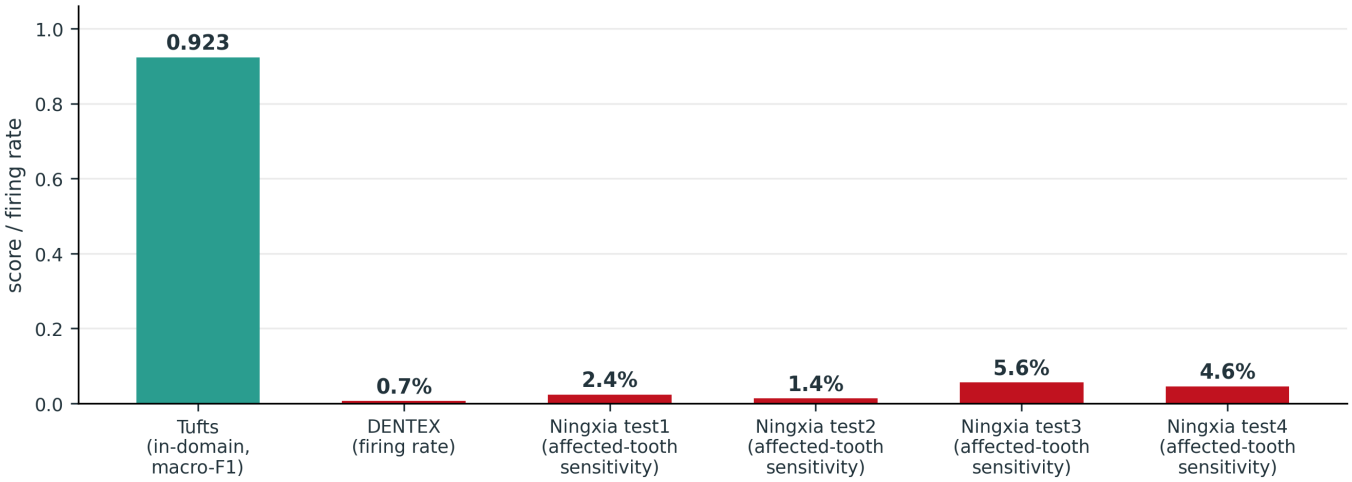



Fig. 5. The bone-loss branch under domain shift. In-domain macro-F1 of 0.923 on Tufts does not transfer: the branch fires on 0.7 % of DENTEX teeth and reaches 1.4–5.6 % affected-tooth sensitivity across the four external hospitals. Metrics necessarily differ between cohorts—the external sets are scored by affected-tooth sensitivity—but the collapse does not depend on that choice. Reported as a measured negative result rather than a component.

answer naming the eight first and second molars is therefore not a degenerate output—it is a compact statement of dental epidemiology, and it is right for the same reason a clinician's prior is right.

The consequence is that headroom above such an answer is small, and—this is the part that matters—small *independently of model quality*. No amount of architectural work raises a ceiling imposed by the pairing of task and metric. A system can only distinguish itself here by being right about *this patient* in a way the population prior is not, and tooth-level set membership is a poor instrument for detecting that difference.

That two unrelated metrics agree on development data is what moves this from anecdote to property. Set-membership F1 and lesion-localisation agreement were constructed to measure different things; the second was adopted precisely because it demands spatial correspondence. A result appearing under one metric is a metric artefact. The same ordering under both is a property of the data.

### B. Anticipated objections

**"Your pipeline is simply weak."** Possibly, but that does not explain the result. Two general-purpose models at 8 B and 33 B parameters were scored by the identical scorer and placed *below* our pipeline, and all three below the constant. An explanation resting on our system's quality would have to account for three systems of very different design failing to separate from the same prior.

**"The constant was fitted, so it is not uninformed."** It is fitted, on 78 held-out images disjoint from every scored image, and then frozen. What it learns is a population statistic—which teeth tend to be carious—not anything about the radiograph in front of it. That is exactly the quantity a diagnostic claim must exceed. We additionally report a test-set-fitted oracle and label it as unclaimable.

**"The metrics were poorly chosen."** The first is the metric this literature predominantly reports; we did not select it to make a point. The second we introduced specifically to escape the first, and it produced the same ordering. We also report the degenerate flag-everything arm, which reaches 0.722 agreement, so the reader can see what the metric costs to saturate and judge its headroom directly.

**"A molar-prior system would still be clinically useful."** It might well be, and we do not dispute it. But a tool that encodes epidemiology is a different object from a tool that reads a radiograph, and only the second can find the lesion that the prior does not expect. The distinction is invisible to the metric and central to the clinic.

### C. Why the external disagreement matters more than the headline

We expected the external cohort to confirm the development result. It did not: pooled, the two systems are indistinguishable, and per site the difference swings through 26 points, from $+15.8$ to $-10.6$.

Two readings remain open and we cannot separate them with the data we have. Either the development result is partly a property of DENTEX's particular caries distribution, or the external metric—which necessarily uses centroid containment, because external lesions are small focal blobs inside much larger tooth boxes—has different sensitivity to the same underlying effect. We report the ambiguity rather than resolve it by assertion.

What the disagreement does establish is firmer than what it leaves open: at this variance, a *single* external number carries very little information. Our own pooled 0.408 is, in isolation, as uninterpretable as any other pooled figure in this literature, including the ones we cite. One hospital in our cohort would have supported the claim that the constant wins decisively;

another would have supported the opposite. Reporting either alone would have been defensible, publishable, and wrong.

### D. The mechanism, so far as we can see it

Our caries gate over-calls healthy teeth at essentially the rate it over-calls impacted teeth. A branch whose false positives are indifferent to what is actually in the crop is one whose decisions carry little information beyond position—which is precisely the regime in which an anatomical prior competes. This was visible in our own error analysis before the constant was ever constructed; the constant quantified what it costs.

### E. What we propose should be reported

**A constant-baseline row.** One line of code, no compute, alongside any tooth-level caries metric on this class of data. Without it a mid-range number cannot be interpreted—our four systems span 0.149 to 0.320 with the *least* informed at the top.

**Per-institution results.** Pooled external figures conceal exactly the variation that external validation exists to expose.

**Metrics with demonstrated headroom.** If a degenerate arm reaches most of the achievable range, the metric cannot support a perception claim regardless of what a system scores on it.

None of this invalidates prior work. It narrows one specific inference: the claim *this system perceives caries* is not established by a tooth-level number alone.

### F. What still separates a system from a prior

A constant cannot name a tooth, cannot express a calibrated confidence, cannot decline informatively, and has no behaviour under domain shift to measure. We propose these as the axes on which OPG systems should be compared, with our own figures given as an illustration of what reporting them looks like:

1) **Identification**—can it say *which* tooth? Here 82.6 % of detected teeth, with the detection bound stated alongside.
2) **Calibration**—are its confidences probabilities? ECE $0.136 \rightarrow 0.043$, with the failed temperature-scaling attempt reported because it diagnoses the miscalibration as structural.
3) **Selective prediction**—does declining improve accuracy, and does the gain survive a shuffle control? $+14.25$ pp, and yes.
4) **Domain shift**—measured on held-out institutions, reported per site.

A constant scores nothing on any of these. Our pipeline scores meaningfully on all four. That asymmetry, rather than its caries F1, is what the system has to show.

### G. On reporting results that do not favour the authors

Two findings here are unfavourable to our own system: the constant comparison and a bone-loss branch that does not generalise. We report both. A screening tool that fails silently is more dangerous than one that fails visibly, and the alternative—selecting whichever metric flatters the system—is precisely the practice this paper argues against. We would have no standing to recommend constant baselines to others while declining to survive one ourselves.

## VII. Limitations

1) **Two caries cohorts, disagreeing in direction.** The constant leads on development data and ties externally. We report both rather than the more striking one; the common finding is non-separability.
2) **No clinician comparison.** We do not know how any system here compares against a dentist on these images. This is the single largest gap and is the subject of ongoing work.
3) **Tooth-level is our finest localisation.** The localisation metric is a tooth-level correspondence check, not a pixel-level segmentation assessment.
4) **One VLM family** at two scales; a domain-adapted medical VLM was not accessible.
5) **Detection bounds Stage 2.** At 0.810 box mAP@0.5, 29.4 % of teeth are never presented to the numbering head.
6) **The external cohort lacks healthy controls**, so specificity and AUC are not computable there; we report lesion agreement only.

## VIII. Conclusion

We began with two programs producing the same report: one that examines a radiograph, and one that has never seen an image in its life. This paper is an attempt to measure the distance between them on the metrics our field actually uses, with our own system as the subject rather than anyone else's.

On development data, the distance was negative. A fixed list of eight molars outscored a purpose-built six-stage pipeline and two vision–language models under two unrelated metrics, and it did so without flagging appreciably more teeth. Across four external hospitals the same comparison produced no separation at all, with a 26-point swing between sites that the pooled figure erases. Meanwhile both language models returned confident, correctly formatted, anatomically specific answers to an image containing no anatomy, and got worse when given examples to learn from.

The conclusion we draw is not that automated caries screening does not work. It is narrower and, we think, more actionable: *this measurement cannot tell us whether it does*. A number that an anatomical prior can reach is not evidence of perception, however carefully it was produced.

What can distinguish a system from a prior is available and inexpensive. Identification, calibration, selective prediction and behaviour under domain shift are all things a constant cannot do at all, and all things we report here with the controls that make them interpretable—a shuffle test for abstention, content-free inputs for the language models, held-out fitting for the baseline. On those axes our pipeline has something to show; on tooth-level caries F1 it does not, and neither, we suspect, do many published figures in the same range.

We therefore recommend three things of the next paper reporting a tooth-level caries result: publish an image-blind

baseline beside it, report external results per institution rather than pooled, and state the headroom of the metric being used. The cost is negligible. The alternative is a literature of numbers that cannot be compared to each other, or to doing nothing at all.

## REFERENCES


[1] Y.-S. Lim, D. Chun, J. Kim, J.-y. Lee, M. J. Ju, J. H. Park, and H.-J. Jeon, "Dual convolutional neural network framework for segmenting dental caries in panoramic radiographs," *Journal of Prosthetic Dentistry*, 2024, article in Press. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0022391323007138

[2] B. Chen, S. Fu, Y. Liu, J. Pan, G. Lu, and Z. Zhang, "CariesXrays: Enhancing caries detection in hospital-scale panoramic dental X-rays via feature pyramid contrastive learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 20. AAAI Press, 2024, pp. 21 940–21 948. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/30197

[3] A. Bilal, A. H. Khan, K. Almohammadi, S. A. Al Ghamdi, H. Long, and H. Malik, "PDCNET: Deep convolutional neural network for classification of periodontal disease using dental radiographs," *IEEE Access*, vol. 12, pp. 10 261–10 275, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10400831

[4] İ. Mutlu and Others, "Comparison of YOLOv11 and YOLOv12 for dental condition detection on panoramic radiographs," *Odontology*, 2026.

[5] K. A. Asif and R. A. Khan, "Dental panoramic radiograph analysis using YOLO26 from tooth detection to disease diagnosis," *arXiv preprint arXiv:2604.16231*, 2026. [Online]. Available: https://arxiv.org/abs/2604.16231

[6] K. Nassiri and M. A. Akhloufi, "YOLO-based panoramic dental X-ray image analysis," *Neural Computing and Applications*, vol. 37, pp. 25 867–25 890, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s00521-025-11462-5

[7] S. Eftimie, T. Ileni, L. Iacob, M. Hedesiu, and L. Diosan, "Tooth-level detection and mapping of dental pathologies on panoramic radiographs using YOLOv11 and RT-DETR," *MethodsX*, vol. 15, no. 3, p. 103696, 2025, published as Sorana Eftimie; an earlier SSRN preprint lists the same author as Sorana Muresanu. Distinct from the unrelated Mureşanu 2024 *Diagnostics* paper already in this corpus (`muresanu2024automating`). [Online]. Available: https://doi.org/10.1016/j.mex.2025.103696

[8] I. D. S. Chen, C.-M. Yang, M.-J. Chen, M.-C. Chen, R.-M. Weng, and C.-H. Yeh, "Deep learning-based recognition of periodontitis and dental caries in dental X-ray images," *Bioengineering*, vol. 10, no. 8, p. 911, 2023. [Online]. Available: https://www.mdpi.com/2306-5354/10/8/911

[9] S. S. Alharbi, A. A. AlRugaibah, H. F. Alhasson, and R. U. Khan, "Detection of cavities from dental panoramic X-ray images using nested U-Net models," *Applied Sciences*, vol. 13, no. 23, p. 12771, 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/23/12771

[10] B. G. Zancan, J. A. Carneiro, C. U. Martins, C. Tirapelli, C. P. Capel, E. D. da Costa, H. Gaêta-Araujo, J. A. Baranauskas, and A. A. Macedo, "AI-powered precision in dental radiographic analysis using tailored CNNs for tooth numbering and cavity detection," *PLOS Digital Health*, vol. 4, no. 2, p. e0000744, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11868773/

[11] Y. Zhang, Y. Yu, Q. Li, H. Chen, Y. Liang, J. Chen, and J. Yin, "A reliable multi-stage system for tooth instance segmentation and numbering in panoramic radiographs," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 3, pp. 2050–2060, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10839580

[12] Z. Meng, J. Hao, X. Dai, Y. Feng, J. Liu, B. Feng, H. Wu, X. Gai, H. Zhu, T. Hu, Y. Wu, H. Xu, J. Li, J. Xiao, X. Liu, J. T. Zhou, F. Zhu, Z. Zhao, L. Xia, B. Fang, J. Sun, J. Wu, and Z. Liu, "A multimodal vision-language model for comprehensive dental diagnosis and enhanced clinical practice," *Nature Communications*, 2026, author list extracted from a fetched summary, not the publisher record directly; volume/issue/pages not resolved at time of writing – verify against the publisher record at proof stage. Preprint precursor available as arXiv:2509.23344 (DentVLM). [Online]. Available: https://doi.org/10.1038/s41467-026-75718-x

[13] D. Jha, T. K. Dutta, R. Paudel, O. K. Susladkar, A. Ghimire, A. Dhakal, S. Adhikari, N. K. Yadav, D. R. Nayak, P. Pawar, W. C. W. Chen, and G. Reynolds, "DentiAsk: A VQA benchmark for multimodal reasoning in panoramic dental radiographs," *arXiv preprint arXiv:2607.08790*, 2026. [Online]. Available: https://arxiv.org/abs/2607.08790

[14] X. Li, X. Liu, C. Liu, S. W. Au Yeung, J. Liu, Y. Yuan, and H. Chen, "PanDent: Toward comprehensive tooth-level structure-language consistency in dental radiology," *arXiv preprint arXiv:2607.27378*, 2026. [Online]. Available: https://arxiv.org/abs/2607.27378

[15] J. Hao, Y. Fan, Y. Sun, K. Guo, L. Lin, J. Yang, Q. Y. H. Ai, L. M. Wong, H. Tang, and K. F. Hung, "Towards better dental AI: A multimodal benchmark and instruction dataset for panoramic x-ray analysis," in *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*. Curran Associates, 2025, also available as arXiv:2509.09254. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/3797b3a3c939156757857cf1b9afde3c-Paper-Datasets_and_Benchmarks_Track.pdf

[16] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, 2020.

[17] J. Cao and Others, "DPML: Deep learning for dental caries detection on panoramic radiographs," *Expert Systems With Applications*, vol. 254, p. 124446, 2024.

[18] D. Cerda Mardini, P. Cerda Mardini, D. Paz Vicuña Iturriaga, and D. R. Ortuño Borroto, "Determining the efficacy of a machine learning model for measuring periodontal bone loss," *BMC Oral Health*, vol. 24, no. 1, p. 1023, 2024. [Online]. Available: https://bmcoralhealth.biomedcentral.com/articles/10.1186/s12903-024-04795-5

[19] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO11," 2024, software, https://github.com/ultralytics/ultralytics.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[21] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[22] K. Panetta, R. Rajendran, A. Ramesh, S. P. Rao, and S. Agaian, "Tufts dental database: A multimodal panoramic x-ray dataset for benchmarking diagnostic systems," *IEEE Journal of Biomedical and Health Informatics*, 2021.

[23] I. E. Hamamci, S. Er, E. Simsar, A. Sekuboyina, and Others, "DENTEX: An abnormal tooth detection with dental enumeration and diagnosis benchmark," 2023, arXiv preprint.

[24] S. Bai, K. Chen, X. Liu, J. Wang, and Others, "Qwen2.5-VL technical report," 2025, arXiv preprint.

[25] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International Conference on Machine Learning (ICML)*, 2017.

[26] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002.